

Only minimal exponents count:

$$Z_{r+2+m}(\sqrt{n}) \sim \left(\prod k_i^{-1} \right) D \frac{C}{2^m m!} n^{-p/2} (\log n)^m \text{ for sorted exponents}$$

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Abstract

Let the candidate exponents $q_i = (h_i + 1)/k_i$ satisfy $q_0 \geq \dots \geq q_r > p = q_{r+1} = \dots = q_{r+1+m}$ (so the minimal exponent p has multiplicity $m+1$). Then the general tower of unit 55 has polylog-bounded, positive levels $G_1, \dots, G_{r+1}$ (unit 56), its level G_{r+2} is an admissible base with mass $C = \int_0^\infty x^{p-q_r-1} G_{r+1} > 0$ and tail rate $(q_r - p)/2$, and $G_{r+2+i} = H_i^{G_{r+2}}$. Consequently $Z_{r+2+m}(\sqrt{n}) \sim (\prod k_i^{-1}) D_{r+2+m} \frac{C}{2^m m!} n^{-p/2} (\log n)^m$, with a complete expansion and power-saving remainder: the multiplicity statement of `thm:TaylorTree` for constant ξ in the chart geometry, for every sorted exponent vector. Zero `sorry/axiom`.

1 Setting

Sorting makes every noncritical step either logarithmic ($q_{i+1} = q_i$) or convergent ($q_{i+1} < q_i$), both of which preserve the polylog-bounded class with $\delta = q_i$; the transition to the minimal block is a convergent step with $t = p - q_r - 1 < -1$, producing a `LogBase`; the minimal block is then unit 49's tower.

2 The things formalised

```

theorem genTower_polyLog ...
  (hsort : i, i < r → chartExp k h (i + 1) chartExp k h i) :
  ( C C : , j : ,
    PolyLogBounded (genTower a k h (r + 1)) C (chartExp k h r) C j)
    x, 0 < x → 0 < genTower a k h (r + 1) x
noncomputable def sortedConst ( a : ) (k h : → ) (r : ) (p : ) : :=
  PolyLogBounded.powLimit (p - chartExp k h r - 1) (genTower a k h (r + 1))
theorem genTower_logBase ... (hp : chartExp k h (r + 1) = p) (hlt : p < chartExp k h r) :
  0 < sortedConst a k h r p M C' : ,
  LogBase (genTower a k h (r + 2)) (sortedConst a k h r p) M C' p
  ((chartExp k h r - p) / 2)
theorem genTower_block_eq_iterDivPrim ... :
  genTower a k h (r + 2 + i) = iterDivPrim (genTower a k h (r + 2)) i
theorem iterChartGen_sorted_isEquivalent ... (hm : 0 < m) ...
  (hsort : i, i < r → chartExp k h (i + 1) chartExp k h i)
  (hp : 1, r + 1 1 → 1 r + 1 + m → chartExp k h 1 = p)
  (hlt : p < chartExp k h r) :

```

```

(fun n => iterChartGen a b k h (r + 2 + m) (Real.sqrt n)) ~[atTop]
  fun n => ( i Finset.range (r + 2 + m), (1 : ) / k i)
    * genScale b k h (r + 2 + m)
    * (sortedConst a k h r p / (2 ^ m * (m.factorial : )))
    * (n ^ (-(p / 2)) * Real.log n ^ m)

```

3 Proof strategy

The class propagation is an induction on r splitting on $q_{r+1} = q_r$ versus $q_{r+1} < q_r$ and applying unit 56's two closure theorems; positivity rides along. The `LogBase` certificate is `logBase_powStep_of_lt` with $t = p - q_r - 1$, after rewriting $t + q_r + 1 = p$ and $(-t - 1)/2 = (q_r - p)/2$. The block identification is the induction of unit 55 with the base level shifted. The asymptotic is the by now standard assembly (units 50, 52) fed with the exact reduction of unit 55 specialised via `prevExp(r + 2 + m) = p`.

4 Roadblocks and resolutions

- Anonymous constructors with placeholder witnesses (`_`, `_`, `_`, `?_`) cannot be elaborated before the proof is known; obtain the fact first and then `exact _`, `_`, `_`, `this` so the witnesses are read off its type.

5 What was Mathlib, what was new

Mathlib: nothing beyond the earlier units. New: the sorted-exponent theorem, its constant, and the expansion.

6 Lessons

Three abstractions (`LogBase` with `rate`, `iterDivPrim`, `PolyLogBounded`) and one exact recursion (`genTower`) reduce the general multiplicity theorem to bookkeeping. The sortedness hypothesis is harmless for the paper (coordinates can be relabelled) but a permutation invariance of the box integral would remove it outright.

7 Outcome

For constant ξ and any exponent vector sorted as above, the chart standard integral has leading term $n^{-p/2}(\log n)^m$ with explicit constant, exactly as `thm:TaylorTree` predicts: half the minimal candidate exponent, and the multiplicity minus one as the power of the logarithm. Together with unit 54 this holds on the paper's domain $(0, b]^d$.